

# DNS Cache Poisoning Lab Report

Lab lab-sample-001 - 2026-09-19T21:12:44 - resolver :5311 - upstream :5301

**Result:** POISONING SUCCESSFUL (3 attempts, strategy known, port randomization OFF, DNSSEC OFF)  
Target bank.test.lab: legitimate 203.0.113.12 vs forged 6.6.6.6

## Cache After Attack

| Domain        | Value        | TTL | Source          |
|---------------|--------------|-----|-----------------|
| test.lab      | 203.0.113.10 | 41  | legitimate      |
| www.test.lab  | 203.0.113.11 | 57  | legitimate      |
| bank.test.lab | 6.6.6.6      | 300 | forged POISONED |

## Resolver Log (key events)

| Time         | Kind      | Detail                                                                  |
|--------------|-----------|-------------------------------------------------------------------------|
| 21:12:20.114 | query     | stub 127.0.0.1:54321 asks bank.test.lab A                               |
| 21:12:20.115 | cache-hit | MISS - forwarding to upstream                                           |
| 21:12:20.116 | forward   | upstream 127.0.0.1:5301 txn=0x9a41 sport=53153                          |
| 21:12:20.140 | reject    | txn ID mismatch (got 0x9a42 expected 0x9a41) - forged response REJECTED |
| 21:12:21.003 | cache-put | bank.test.lab -> 6.6.6.6 cached (TTL 300, forged accepted)              |
| 21:12:21.004 | reject    | late legitimate reply discarded (socket already closed)                 |

## Defensive Recommendations

DNSSEC validation is the definitive mitigation - forged responses lack valid RRSIG signatures and are rejected. Source port randomization raises the attacker's search space past a billion attempts. Deploy bailiwick checking on recursive resolvers. This lab ran on an isolated network only - never aim the tool at production DNS.